Arista Access Node - HEB-ACC-51: Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) Troubleshooting SOP

Node: HEB-ACC-51 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9218

Description

The Routing Protocol Flap failure domain on Arista Access Node HEB-ACC-51 at the Hebbal Edge site can be attributed to the high volatility of Routing Information Protocol (RIP) variants like BGP, OSPF, and ISIS LSP flooding, which are intrinsic to the Access role and the Arista vendor equipment. The BGP route churn occurs due to the dynamic nature of BGP's route advertisement and withdrawal, triggered by various events such as network topology changes or misconfigured BGP neighbors. In contrast, OSPF adjacency loss is typically a result of OSPF neighbor discovery issues or OSPF LSA mismatch. Similarly, ISIS LSP flooding is a common occurrence in the presence of ISIS LSP mismatch or adjacency issues. The root cause analysis of this failure domain necessitates a thorough examination of the underlying protocol or hardware mechanics, including network topology, routing tables, and BGP/OSPF/ISIS configurations.

Pre-requisite Checks

- Prerequisite: Verify that the Hebbal Edge site is accessible and that all necessary physical checks have been performed.

- Prerequisite: Check the status of redundant links and ensure that they are functioning correctly.

- Prerequisite: Dispatch remote hands to the Hebbal Edge site to perform any necessary hardware checks or adjustments.

Potential Root Causes

Cause	Probability
BGP route churn due to network topology changes or misconfigured BGP neighbors	High
OSPF adjacency loss due to OSPF neighbor discovery issues or OSPF LSA mismatch	Medium
ISIS LSP flooding due to ISIS LSP mismatch or adjacency issues	Low

Diagnostic Commands & Expected Outputs

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Command: show ip bgp summary, Expected Output: BGP table version is 1, main routing table version is 1, Description: This command displays a summary of the BGP routing table, including the BGP table version and the main routing table version. This allows for the verification of BGP route churn.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays
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information about OSPF neighbors, including the neighbor ID, priority, state, dead time, address, and interface. This allows for the verification of OSPF adjacency loss.

Command: show isis database, Expected Output: ISIS LSP Database, Description: This command displays the ISIS LSP database, including information about ISIS LSPs and their attributes. This allows for the verification of ISIS LSP flooding.
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Resolution Steps

Step 1: Step: Verify and adjust BGP configurations as necessary, Command: no bgp best path as-path multipath-relax, Description: This command removes the 'multipath-relax' keyword from the BGP best path configuration, which can help resolve BGP route churn issues.

Step 2: Step: Verify and adjust OSPF configurations as necessary, Command: no ip ospf dead-interval, Description: This command removes the dead interval configuration from the OSPF protocol, which can help resolve OSPF adjacency loss issues.

Step 3: Step: Verify and adjust ISIS configurations as necessary, Command: no isis net, Description: This command removes the ISIS network configuration, which can help resolve ISIS LSP flooding issues.

Post-Check Verification

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Command: show ip bgp summary, Expected Output: BGP table version is 0, main routing table version is 0, Description: This command verifies that BGP route churn has been resolved.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command verifies that OSPF adjacency loss has been resolved.

Command: show isis database, Expected Output: ISIS LSP Database, Description: This command verifies that ISIS LSP flooding has been resolved.
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Roll-back Procedures

- Step: Reapply the original BGP configuration, Command: bgp best path as-path multipath-relax, Description: This command reapplies the original BGP best path configuration, which can help resolve any issues caused by the previous resolution step.

- Step: Reapply the original OSPF configuration, Command: ip ospf dead-interval, Description: This command reapplies the original OSPF dead interval configuration, which can help resolve any issues caused by the previous resolution step.

- Step: Reapply the original ISIS configuration, Command: isis net, Description: This command reapplies the original ISIS network configuration, which can help resolve any issues caused by the previous resolution step.